

RESEARCH PAPER ON “AAKRITI”

A platform that converts 2D radiology reports (X-ray, CT, MRI) into interactive 3D/AR visuals to improve patient understanding

ABSTRACT :

“AAKRITI: An AI-Driven 2D-to-3D Augmented Reality Platform for Patient-Centric Visualization of Medical Imaging in India”

AAKRITI is a healthcare platform that transforms 2D radiological images (X-ray, CT, MRI) into interactive 3D models and augmented reality (AR) experiences to support patient education and communication in Indian clinical settings. Existing literature shows that AR improves visualization of complex anatomical structures and enhances patient understanding during education interventions. Concurrently, AI-based 2D-to-3D reconstruction methods enable accurate 3D models from standard radiographs, offering advanced visualization without additional imaging. India's acute radiologist shortage and rising imaging volumes highlight the need for tools that streamline consultations and communication. This paper presents the design, feasibility and potential impact of AAKRITI, targeting doctors, hospitals, clinics and diagnostic centers as a service-based application integrated into routine radiology workflows

INTRODUCTION :

Augmented reality (AR) has gained increasing attention in medical imaging as a means to enhance visualization of complex anatomical structures and improve communication among healthcare professionals. By overlaying medical imaging data onto the real-world environment, AR can provide an intuitive spatial understanding that complements conventional 2D screens. Recent studies show that AR and extended reality systems are effective for medical education, surgical planning and image-guided procedures, indicating their potential to transform diagnostic imaging workflows.

In parallel, AR has been systematically studied for patient education, where interactive 3D content improves perceived knowledge gain, information recall and engagement compared to traditional leaflets or verbal explanations alone. AI-driven 2D-to-3D reconstruction techniques now enable clinically useful 3D models from standard radiographic images, reducing reliance on additional CT scans and making advanced visualization more accessible.

In India, radiology services are under pressure from rapid growth in imaging volumes and a persistent shortage of radiologists. Estimates suggest that India has roughly one radiologist per 100,000 people, with imaging volumes increasing by 15–20% annually, resulting in heavy workloads and limited time for detailed patient counselling. In this context, tools that streamline doctor–patient communication and efficiently convey imaging findings are increasingly valuable.

AAKRITI is proposed as an AI-enabled AR platform that converts conventional 2D radiology images into interactive 3D and AR visualizations delivered at the point of care. It targets doctors, hospitals, clinics and diagnostic centers in Nagpur and across India, providing an additional patient-education layer as a service integrated into existing imaging workflows.

PROBLEM STATEMENT :

Clinical communication gap:

Radiology findings are often communicated to patients using printed reports and static 2D images, which are difficult for non-experts to interpret. As a result, patients may not fully understand the nature or severity of their condition, leading to confusion, anxiety and suboptimal shared decision-making.

Time pressure on radiologists and clinicians in India:

In India there is a well-documented shortage of radiologists relative to the growing demand for imaging, with estimates of only about one radiologist per 100,000 people. Some industry analyses describe an even higher workload, equating to approximately one radiologist for every 65,000 individuals as imaging volumes grow by 15–20% per year. This imbalance constrains the time available for radiologists and referring clinicians to explain imaging results in detail to each patient.

Limited use of advanced visualization for routine patient education:

Although 3D reconstruction and AR visualization have demonstrated benefits in surgical planning, medical education and complex case review, they are rarely deployed at scale for everyday patient consultations due to cost, technical complexity and workflow integration barriers. Existing systems often require specialized hardware or dedicated workstations, making them less accessible for smaller clinics and diagnostic centers.

Resulting problem:

There is a need for an affordable, workflow-compatible solution that transforms routine 2D radiology images (X-ray, CT, MRI) into intuitive 3D and AR visualizations, enabling clinicians in India to explain findings quickly and clearly, improving patient understanding while respecting time and resource constraints.

TARGETED USERS :

AAKRITI is designed for multi-stakeholder use within the Indian healthcare ecosystem:

Primary users – Clinicians and radiologists:

Radiologists and referring doctors use the platform to convert 2D imaging studies into interactive 3D/AR visualizations during consultations, improving communication and shortening explanation time.

Institutional customers – Hospitals, clinics, diagnostic centers:

Hospitals and diagnostic centers integrate AAKRITI into radiology workflows as a value-added service that differentiates their offerings, supports patient-centric care and may improve satisfaction scores.

Indirect beneficiaries – Patients and caregivers:

Patients and their families receive AR-based explanations of imaging findings, which have been associated in prior AR education studies with higher perceived knowledge and better information recall. AAKRITI can be delivered chair-side or via mobile devices, aligning with increasing smartphone penetration in India.

MARKET SIZE :

The Indian diagnostic imaging ecosystem represents a substantial and growing opportunity for solutions integrated into X-ray, CT and MRI workflows. Recent market analyses estimate that the India diagnostic imaging services market is valued at approximately USD 6.33 billion in 2024 and is projected to reach about USD 11.55 billion by 2032, with a compound annual growth rate (CAGR) of around 7.8%.

In parallel, the India augmented reality (AR) in healthcare market is estimated at roughly USD 81.53 million in 2024 and forecast to reach USD 529.53 million by 2033, growing at an estimated CAGR of 21.4% from 2025 to 2033.

AAKRITI operates at the intersection of these two domains, leveraging AR as a visualization layer on top of conventional diagnostic imaging services; therefore, its total addressable market (TAM) can be framed as the combined space of diagnostic imaging services and AR-enabled healthcare applications in India.

For the serviceable available market (SAM), the scope is narrowed to imaging workflows where 2D-to-3D visualization and AR-based patient education are directly applicable—specifically, X-ray, CT and MRI-based diagnostic imaging performed in hospitals, clinics and diagnostic centers. Industry data indicate that X-ray services account for approximately 26.5% of the diagnostic imaging services market in India, reflecting their use as a first-line modality, while CT and MRI segments are also expanding steadily.

The India portable X-ray and CT devices market alone generated about USD 126.9 million in 2024, with projections to reach USD 177.7 million by 2030, and the India CT scanners market is estimated at USD 303.77 million in 2024, projected to reach USD 592.74 million by 2033. These figures illustrate the growing installed base of imaging equipment and the increasing volume of examinations that could potentially incorporate AAKRITI as a visualization and communication layer, defining a sizable SAM within the broader multi-billion-dollar imaging services market.

The serviceable obtainable market (SOM) for AAKRITI is necessarily more conservative and reflects realistic early adoption focused on Nagpur and selected Indian cities. In an initial three-to-five-year horizon, it is reasonable to model SOM as a small fraction of the SAM—on the order of 0.5–1%—corresponding to a limited but meaningful share of the imaging studies conducted in partner hospitals, clinics and diagnostic chains. This scenario assumes gradual onboarding of institutions that are early adopters of AR-enabled patient-centric imaging services and is aligned with the observed growth trajectories of both the diagnostic imaging services market (CAGR ~7–8%) and the AR-in-healthcare segment (CAGR ~21%) in India. These values should be presented as illustrative adoption scenarios rather than exact forecasts, but they highlight the commercial potential of an AI- and AR-enhanced visualization platform like AAKRITI within the Indian healthcare context.

Metric	Definition (for AAKRITI)	Example Scope / Values
TAM	Total addressable market: full potential spend in domains where AAKRITI's technology could be applied.	India diagnostic imaging services market (~USD 6.33B in 2024, projected ~USD 11.55B by 2032) plus India AR in healthcare market (~USD 81.53M in 2024, projected ~USD 529.53M by 2033). databridgemarketresearch+1
SAM	Serviceable available market: segment where AAKRITI can practically be used.	Subset of the diagnostic imaging services market corresponding to X-ray, CT and MRI workflows in hospitals, clinics and diagnostic centers; X-ray ~26.5% of imaging services, with growing CT/MRI and portable imaging markets. databridgemarketresearch+2
SOM	Serviceable obtainable market: realistic share AAKRITI can capture in the initial years.	Early adoption in Nagpur and selected Indian cities; illustrative scenario of ~0.5–1% of SAM over 3–5 years, contingent on institutional partnerships and AR adoption rates. databridgemarketresearch+2

COMPETITOR ANALYSIS :

Category	Representative Solutions / Examples	Strengths	Limitations vs. AAKRITI's Focus
Medical 3D visualization software	3D Slicer, Materialise Mimics, Simpleware, OsiriX MD, RadiAnt, 3D-DOCTOR alternatives. encord+1	Advanced 3D reconstruction, segmentation, surgical planning, DICOM support; widely used in research and surgery planning. encord+1	Desktop/workstation-centric, require expert users and training; oriented to planning and research rather than brief patient explanations in routine radiology; often costly for smaller centers. encord+2
AR-based patient education apps	Smartphone-based AR radiotherapy education app showing patients a 3D simulation of their own treatment plan.	Demonstrated improvement in patient understanding and reduced anxiety; runs on standard smartphones/tablets without special hardware.	Focused on a specific treatment domain (radiotherapy); not generalized as a scalable product for diverse radiology exams and everyday diagnostic workflows in India. pmc.ncbi.nlm.nih+1
AR/VR imaging & anatomy visualization platforms	AR/VR systems for interactive 3D anatomy (e.g., HoloView) and advanced visualization in radiology using 3D, VR and AR. pmc.ncbi.nlm.nih+2	Immersive visualization, gesture-based interaction, strong fit for education and complex case review; demonstrate feasibility of AR in imaging. pmc.ncbi.nlm.nih+2	Often research prototypes or specialized platforms requiring dedicated hardware and integration; oriented to education and tertiary centers rather than high-throughput diagnostic centers and simple patient communication. pmc.ncbi.nlm.nih+2

UNIQUE VALUE PROPOSITION :

AAKRITI provides a lightweight, smartphone-compatible layer on top of routine X-ray, CT and MRI workflows, enabling clinicians in Indian hospitals, clinics and diagnostic centres to convert standard 2D images into intuitive 3D and augmented reality (AR) views during everyday consultations.

Unlike workstation-centric 3D visualization software and head-mounted AR systems that require specialized hardware, expert users and significant investment, AAKRITI is designed for low-friction deployment using existing radiology infrastructure and commodity mobile devices, making it suitable for high-volume diagnostic environments in India. By combining AI-based 2D-to-3D reconstruction with AR-based patient education, AAKRITI helps radiologists and referring doctors explain complex imaging findings quickly and clearly, improving patient understanding and experience while saving clinician time in settings already affected by radiologist shortages and heavy workloads.

BUSINESS MODEL :

AAKRITI follows a B2B and B2B2C model, selling primarily to hospitals, clinics and diagnostic centres while ultimately benefiting patients through better imaging communication. Healthcare providers integrate AAKRITI into existing X-ray, CT and MRI workflows to convert routine 2D studies into 3D and AR visualizations shown during consultations. The platform is offered as cloud-based (or on-premise) software accessible via web and mobile apps, minimizing upfront hardware costs and following the growing SaaS-style preference in Indian healthcare IT. Operationally, AAKRITI functions as a plug-in layer on top of existing PACS/RIS and imaging equipment, deployable on commodity smartphones or tablets with minimal workflow disruption. This software-only approach makes the solution scalable across multiple centres, enabling diagnostic chains to standardize patient-education experiences while centrally managing configuration, updates and analytics.

REVENUE MODEL :

AAKRITI uses a tiered SaaS subscription model, optionally complemented by per-use or value-based components. Diagnostic centres and hospitals pay a recurring monthly or annual fee linked to factors such as number of sites, active clinician users or approximate imaging study volume, similar to EMR and practice-management software pricing commonly used in Indian healthcare. Tiered plans (e.g., basic, standard, premium) can differentiate functionality, including access to advanced AI-based 2D-to-3D features and analytics dashboards. In addition, providers can bundle AAKRITI as a value-added service within their imaging packages, indirectly monetizing it by improving patient experience and strengthening patient retention rather than billing it as a separate line item. For larger diagnostic chains, enterprise contracts with multi-site licensing, SLAs, training and customization offer an additional revenue stream, while the core model remains centred on predictable subscription income.

“For example, a basic plan could be priced around INR 5,000–10,000 per month per centre, in line with entry-level EMR subscriptions in India, while higher tiers for multi-site chains could reach INR 50,000+ per month depending on volume and features.”

FEASIBILITY ANALYSIS :

Technical feasibility.

AAKRITI builds on technologies that have already been demonstrated as technically feasible in clinical and near-clinical settings. Smartphone-based AR applications have been successfully developed for patient education using portable iOS and Android devices, allowing patients to view personalized 3D simulations and demonstrating good usability and comprehension. Studies comparing smartphone-based and head-mounted AR platforms in interventional radiology show that smartphone AR can provide accurate image overlay and navigation guidance, with improved task performance relative to conventional freehand techniques. More broadly, reviews of smartphone-based imaging systems in healthcare conclude that modern mobile devices have sufficient camera, processing and graphics capabilities to support advanced visualization and image-guided applications at low cost. In parallel, AI-based 2D-to-3D reconstruction methods for medical imaging are maturing, enabling the generation of clinically meaningful 3D models from standard radiographs or CT slices. Together, this evidence indicates that the core building blocks for AAKRITI—AI-driven 2D-to-3D conversion and smartphone-based AR visualization—are technically feasible using current hardware and software stacks.

Operational feasibility.

From an operational standpoint, AAKRITI is designed to function as an add-on layer to existing radiology workflows rather than a replacement for PACS/RIS or reporting systems, which reduces disruption risk. This aligns with broader trends in healthcare SaaS adoption in India, where clinics and hospitals seek cloud-based tools that can be integrated into current processes with minimal implementation overhead. Reports on diagnostic centres and digital transformation highlight that technology is increasingly used to improve process efficiency, standardize quality and enhance patient experience across distributed sites, especially when deployed through centralized platforms. By running on commodity smartphones and tablets already in use by clinicians, and by requiring only incremental changes to how X-ray, CT and MRI images are accessed and presented during consultations, AAKRITI's workflow impact is limited and can be managed through short training sessions and phased rollouts. This suggests that, given appropriate change management and stakeholder engagement, operational adoption in Indian diagnostic centres and hospitals is feasible.

Financial feasibility.

Financially, AAKRITI leverages a SaaS-based model that aligns with the constraints of Indian healthcare providers. Analyses of healthcare SaaS and patient-portal solutions in India emphasize that subscription-based pricing with low upfront capital expenditure makes digital tools more accessible to both small clinics and larger hospital chains. Pre-feasibility reports for diagnostic centres also highlight the role of private partners in investing in technology to improve service quality and efficiency while operating within government-approved or competitive market rates. Because AAKRITI is delivered as software using existing imaging equipment and consumer-grade mobile devices, its marginal deployment cost per centre is relatively low compared to new scanners or infrastructure upgrades. Revenue can be generated through recurring subscriptions and optional per-scan fees, with AR-enhanced explanations positioned as a value-added service that supports patient satisfaction and differentiation rather than a major cost driver. Taken together, this suggests that a carefully priced, SaaS-based AR visualization platform like AAKRITI is financially feasible in the Indian diagnostic imaging market, especially when rolled out incrementally and bundled with demonstrable improvements in patient communication and experience.

SCALABILITY :

AAKRITI is designed to scale horizontally across facilities and geographies by leveraging a cloud-based SaaS architecture and commodity mobile hardware. Cloud-based medical imaging and healthcare SaaS solutions are recognized as key enablers of scalability, allowing providers to accommodate rising imaging volumes and user numbers without proportional increases in on-premise infrastructure. By deploying AAKRITI as a centrally managed service with web and smartphone clients, new hospitals, clinics and diagnostic centres can be onboarded with limited local installation effort, while software updates, security patches and feature enhancements are rolled out uniformly from the cloud. This model is particularly suited to India, where diagnostic chains are rapidly expanding their laboratory and imaging networks across multiple cities and states to meet growing demand for preventive and advanced testing.

At the technical level, best practices for scalable healthcare software—such as modular architecture, API-first integration and cloud-native deployment—enable AAKRITI to interface with heterogeneous PACS/RIS and hospital information systems while maintaining performance under increasing load. A modular design allows new AI models, visualization features or analytics modules to be added without disrupting existing services, and API-driven interoperability supports integration with partner platforms, including AI-radiology solutions and patient-portal systems. On the demand side, scalability is reinforced by the fact that the platform runs on widely available smartphones and tablets rather than specialized AR hardware, which lowers barriers to adoption for smaller centres and enables consistent deployment across large networks of diagnostic sites. As a result, once validated in pilot locations, AAKRITI can be rolled out incrementally across multi-centre diagnostic chains and hospital groups, supporting a growing number of clinicians and patients without fundamental changes to the core architecture.

Go-To-Market Strategy :

The go-to-market (GTM) strategy for AAKRITI focuses on a phased B2B rollout, beginning with early-adopter diagnostic centres and hospitals in Nagpur and subsequently expanding to other Indian cities. Reports on diagnostic services in India highlight rapid growth driven by rising imaging demand, increasing corporate chains and strong competition on service quality. In this environment, AAKRITI is positioned as a differentiation tool that enhances patient communication and experience without requiring additional imaging equipment, aligning with digital-health GTM frameworks that emphasize clear clinical value, workflow fit and low implementation friction.

Phase 1 – Pilot deployments in Nagpur.

The initial phase targets a small number of mid- to large-sized hospitals and diagnostic centres in Nagpur that already have digital radiology workflows and are open to innovation. GTM literature for diagnostic centres recommends starting with focused pilots to demonstrate measurable benefits—such as improved patient satisfaction scores, reduced explanation time and positive feedback from clinicians—before broader scaling. In this phase, AAKRITI can be offered on favourable pilot terms, accompanied by on-site onboarding, short training sessions and co-branded patient-education materials, generating early case studies and reference sites.

Phase 2 – Expansion to regional and national diagnostic chains.

After validating product-market fit and clinical acceptance, AAKRITI's second phase focuses on partnering with regional and national diagnostic chains, which are increasingly central to India's diagnostics market. Market analyses indicate that such chains prioritize standardization of quality, centralized IT platforms and differentiated patient experience across multiple locations. AAKRITI's SaaS architecture allows multi-site deployments with centralized administration, making it suitable for chain-level contracts that bundle licensing, training and support under enterprise agreements. Collaboration with AI-radiology equipment suppliers and health-tech integrators can further accelerate penetration by embedding AAKRITI into broader digital imaging solutions promoted to hospitals and centres.

Phase 3 – Ecosystem integration and digital marketing.

In the longer term, the GTM strategy includes deeper integration into the broader digital health ecosystem and targeted marketing to both clinicians and patients. Digital-transformation reports for Indian healthcare underline the importance of interoperable solutions that integrate with hospital information systems, patient portals and telemedicine platforms. AAKRITI can offer APIs and integration options for existing HIS/PACS vendors, expanding reach through partnerships rather than standalone sales. On the demand side, health-tech GTM frameworks recommend using thought-leadership content, clinician-led webinars and case reports to build trust among radiologists and referring doctors, combined with patient-facing campaigns that highlight clearer explanations and better understanding of imaging results. By combining pilot-led validation, chain-level partnerships and ecosystem integrations, AAKRITI's GTM strategy is designed to scale sustainably within the evolving AI-assisted radiology market in India.

SWOT ANALYSIS:

Strengths

Uses proven technologies: AI-based 2D-to-3D reconstruction and smartphone-based AR are already demonstrated as feasible and effective for medical visualization and patient education.

Patient-centric and time-saving: AR explanations can improve patient understanding and satisfaction while reducing explanation time for clinicians.

Low hardware barrier: Runs on existing imaging infrastructure and commodity smartphones/tablets, avoiding expensive AR headsets and specialized workstations.

SaaS model: Cloud-based architecture supports easy updates, multi-centre deployment and lower upfront costs for hospitals and diagnostic centres.

Weaknesses

Dependence on image quality and data integration: Performance relies on access to good-quality DICOM images and smooth integration with PACS/RIS systems, which may vary between centres.

Need for clinician adoption and training: Radiologists and clinicians must change their routine to use AR during consultations, requiring training and behaviour change.

Limited initial validation: As a new solution, AAKRITI will initially have limited clinical evidence and few reference sites compared with established 3D/AR platforms.

Opportunities

Growing imaging and AI-radiology market: India's diagnostic and AI-assisted radiology markets are expanding, creating demand for tools that enhance efficiency and communication.

Rising digital health adoption: Policy support and investment in India's digital health ecosystem favour scalable SaaS solutions that improve patient experience and workflow.

Differentiation for diagnostic chains: Large diagnostic chains seek ways to differentiate on service quality and patient experience; AR-enhanced explanations can be a visible value-added service.

Extension to education and telehealth: The same platform can later support medical education, remote counselling and telemedicine use-cases using AR visualizations.

Threats

Competitive 3D/AR and AI-radiology solutions: Other vendors may offer integrated AI-radiology and visualization platforms, including global companies with strong brand and distribution.

Regulatory and data-privacy requirements: Evolving regulations on health data, AI and medical devices in India may impose compliance burdens and slow deployment.

Resistance to workflow change: Some clinicians and administrators may be reluctant to adopt new tools if they perceive them as adding steps or time, especially in high-volume centres.

Economic constraints: Smaller centres with tight margins may be slow to pay for additional SaaS tools unless the financial and clinical benefits are clearly demonstrated.

RISK AND CHALLENGES :

Regulatory and data-privacy risk.

AAKRITI works with sensitive medical images and therefore must comply with evolving Indian regulations on health data, AI and digital privacy (e.g., DPDP Act and digital health guidelines). Reviews of AI in radiology highlight regulatory compliance, cybersecurity and data-privacy protection as major barriers to adoption, with breaches or misuse potentially eroding patient trust and exposing providers to legal and reputational risk.

Workflow integration and adoption challenges.

Experience with AI and AR tools in radiology shows that lack of seamless workflow integration is a key reason many solutions fail to scale beyond pilots. Hospitals often struggle with under-resourced IT teams, siloed data and limited interoperability between PACS, RIS and electronic health records; each additional system that clinicians must log into or manage can increase friction and resistance to change. If AAKRITI is not tightly integrated into existing image-viewing and reporting workflows, clinicians may perceive it as extra work rather than a time-saving tool.

Technical limitations of AR and AI.

Current AR and AI technologies still have limitations that can affect clinical usability. Studies on AR in clinical settings report challenges related to image quality, latency, field of view and alignment accuracy, which can reduce user confidence if not carefully addressed. Similarly, AI models for 2D-to-3D reconstruction may be affected by training-data bias or fail in edge cases, and radiology AI literature warns about errors, lack of robustness and difficulty in explaining model decisions. Ensuring consistent performance across different scanners, protocols and centres is a non-trivial technical challenge.

Economic and adoption risk in Indian healthcare.

Digital-health analyses for India note that many hospitals and diagnostic centres operate under tight financial constraints and may be cautious about paying for new SaaS tools unless there is clear evidence of clinical and financial benefit. AI and AR solutions in radiology face uncertainty regarding return on investment, especially when the benefits are indirect (e.g., patient satisfaction, brand differentiation) rather than direct revenue. Smaller centres might delay adoption or choose lower-cost competitors, and broader macro-economic or reimbursement changes could also slow investment in value-added technologies like AAKRITI.

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